Torchvision Environment Setup

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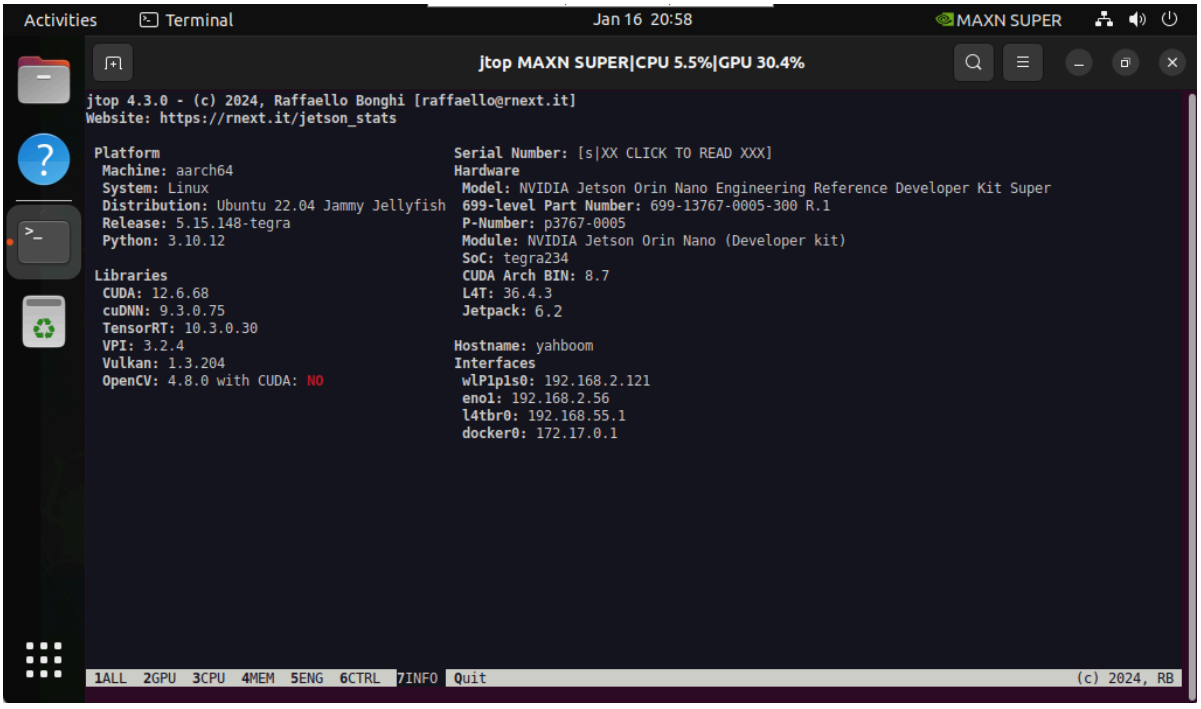
References

> **Note on pip installation:** Ubuntu 24.04 marks the system Python as "externally managed" according to PEP 668, so installing packages directly with pip may report `externally-managed-environment`. The `--break-system-packages` option allows pip to install packages in that environment; virtual environments usually do not need this option, but it is kept here to keep the commands in this tutorial consistent.

> The factory system already has the relevant environment set up, so the installation steps can be skipped.

Torchvision provides computer vision datasets, models, and image transforms for PyTorch.

1. System Information



The current environment uses PyTorch `2.8.0`, Torchvision `0.23.0`, and Python `3.12`. Torchvision `0.23.0` corresponds to PyTorch `2.8.0`.

2. Installing Dependencies

```
sudo apt update
sudo apt install ninja-build libwebp-dev libjpeg-dev -y
```

3. Installing Torchvision

According to the version correspondence between PyTorch and Torchvision, PyTorch `2.8` uses Torchvision `0.23`, which supports Python `3.9` through `3.13`. When installing, use the version that fits Jetson ARM64, Python 3.12, and the current PyTorch.

3.1. Offline Installation

Prepare in advance the Torchvision `0.23.0` wheel file that matches the current environment, and copy it to the Jetson. Do not use the `0.20.0a0` / `cp310` wheels from the old tutorials.

```
cd Downloads/
python3 -m pip install --user --break-system-packages ./<Torchvision-0.23.0 ARM64 wheel file name>
```

3.2. Online Installation

On a device with internet access, install from the PyTorch Vision `v0.23.0` source. This method uses the currently installed PyTorch and does not automatically replace its dependencies:

```
python3 -m pip install --user --break-system-packages --no-deps --no-build-isolation "git+https://github.com/pytorch/vision.git@v0.23.0"
```

3.3. Building from Source

Download the `v0.23.0` source code:

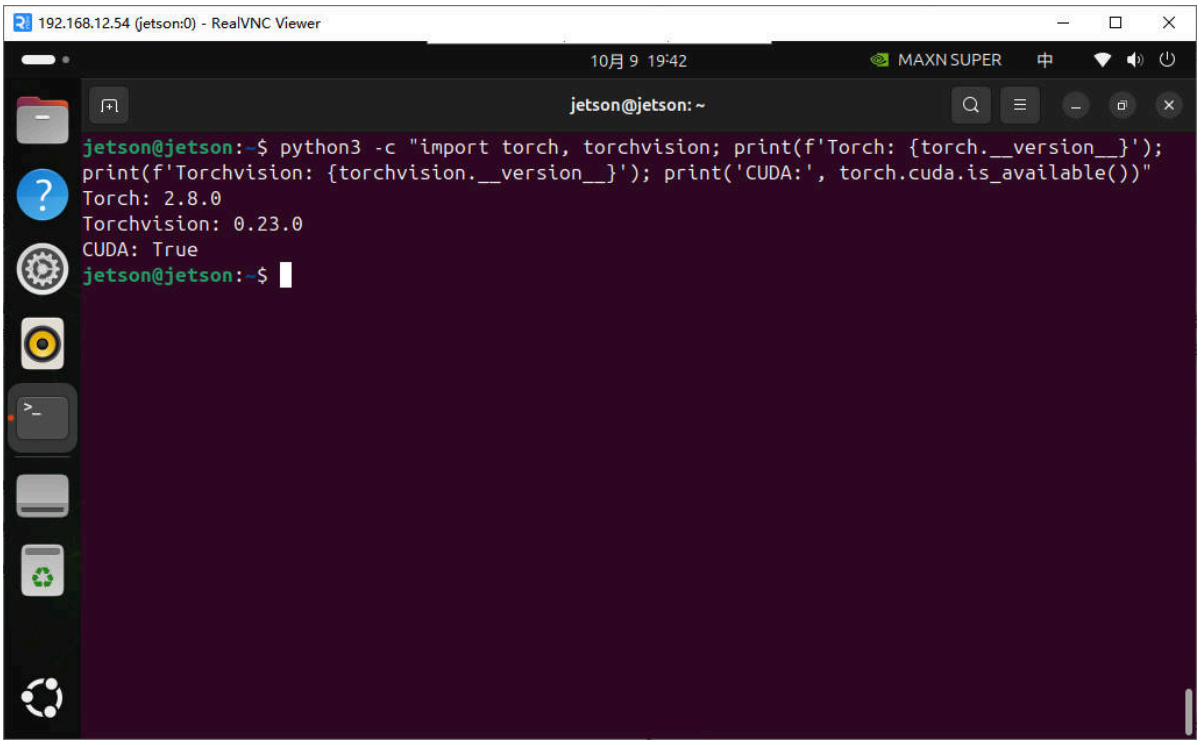
https://github.com/pytorch/vision/tree/v0.23.0

Enter the source code directory and install:

```
cd vision-0.23.0
python3 -m pip install --user --break-system-packages --no-deps --no-build-isolation .
```

4. Verifying the Installation

```
python3 -c "import torch, torchvision; print(f'Torch: {torch.__version__}'); print(f'Torchvision: {torchvision.__version__}'); print('CUDA:', torch.cuda.is_available())"
```



References

https://github.com/pytorch/vision

https://docs.ultralytics.com/guides/nvidia-jetson/